

ESAT Physics Revision Guide

ENGAA 2017 — Multiple-Choice Physics

A full lecture-replacement worked from the tutorial board and audio

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Questions: Q1, Q2, Q4, Q6, Q8, Q10, Q12, Q14

“This was a good first ESAT preparation session. The aim is not just to get answers, but to build the habit of recognising the physics quickly and choosing the most efficient route.”

— **Tom Rogerson**

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1 ESAT Exam Strategy — Tom’s Coaching

This first section captures *every* piece of exam-technique coaching Tom gave. The ESAT is not primarily a harder-physics test than A-level — it is a **time-pressure and interpretation** test. You have roughly **27 questions in 40 minutes** (≈ 89 seconds each), **no calculator**, and **no formula sheet**. Everything below is built around that reality.

The headline principle: “Bad habits for A-level are good habits for ESAT”

At A-level you are rewarded for rigour: full working, exact values, every step shown. The ESAT rewards the opposite instincts:

- **Skip steps.** If you can jump straight from the physics idea to the number, do it. Don’t write out the middle algebra.
- **Approximate everything.** Round aggressively, use nice numbers, and only chase precision if two answer options are close.
- **Use $g = 10 \text{ N/kg}$, never 9.81.** The paper itself gives $g = 10 \text{ N kg}^{-1}$ — so the arithmetic is always meant to be clean.
- **Read the magnitude of the answer options first.** They tell you what scale the answer should be, and let you sanity-check instantly.

The rigour that earns marks in a written A-level exam *costs* you time here. Retrain the instinct.

Identify the physics first, then compute

Tom’s single most repeated instruction. Before touching any numbers, ask: “*What is this question actually about?*” Name the topic (convection, energy conservation, path difference, nuclear balance, ...), write down the one key idea, and only *then* decide the quickest route to the number. Most ENGAA physics questions have a short route once the context is understood — the difficulty is dressed up in the wording, not in the maths.

Don’t go in order — bank the easy questions

You are not obliged to work through the paper top to bottom. In this session you deliberately worked the opening maths warm-up and then jumped to the physics questions rather than grinding through every question in sequence. Tom praised this explicitly:

“This is a sensible approach for ESAT preparation because the aim is not just to get answers, but to build the habit of recognising the physics quickly and choosing the most efficient route.”

Strategy: scan the paper, pick off the questions you can do fast, bank those marks, and leave the time-sinks for a second pass. A marked question you can definitely do is worth more than a hard one you *might* do.

No calculator — train your mental maths

Every value on this paper is chosen so the arithmetic is doable in your head: $1200 \times 10 \times 1 = 12000$, $12000/0.8 = 15000$, $3 \times 10^8 \times 4 \times 10^{-10} = 12 \times 10^{-2}$. The exam is testing whether you can *organise* a calculation, not whether you own a calculator. Practise:

- Standard-form multiplication (multiply the digits, add the powers of ten).
- Dividing by efficiencies given as “nice” fractions ($\div 0.8 = \times 1.25$, $\div 0.75 = \times \frac{4}{3}$).

- Reading powers of ten off the answer options to fix the exponent.

Priority order for every question

Tom's summary of what to become confident at, in order:

1. **Spot the topic quickly** — what branch of physics is this?
2. **Decide which information actually matters** — ENGAA questions deliberately include numbers you don't need (e.g. the detector separation in Q8). Filter them out.
3. **Avoid long methods** when a simple physical argument or a proportional shortcut will do the job.

The timeline — October or January, either way you win

Tom framed the bigger picture clearly:

“If you're ready in October, great. If not, it's still good practice for January.”

Practising these ENGAA questions is good maths and good physics regardless of which sitting you take. There is no wasted effort: the same skills feed straight into your A-level Unit 1 work. The goal for now is simply to **build speed** and to keep sharpening the “recognise → shortcut” reflex.

2 Prior-Knowledge Bootcamp — First Principles

Before the question-by-question walkthrough, here is every physics concept the lesson relied on, built up from first principles. If you can explain each of these out loud without notes, the questions become fast.

2.1 Density and Thermal Expansion

Density

Density is how much mass is packed into a given volume:

$$\rho = \frac{m}{V}$$

where ρ is density (kg m^{-3}), m is mass (kg), V is volume (m^3). For a *fixed mass* of material, density and volume are inversely related: if V goes up and m is unchanged, ρ must go down.

Thermal expansion. When you heat almost any substance, its particles gain kinetic energy, vibrate/move more, and push each other slightly further apart on average. The mass is unchanged (you have not added or removed particles) but the volume increases. Therefore:

$$\text{heat} \longrightarrow \text{particles spread out} \longrightarrow V \uparrow \longrightarrow \rho = \frac{m}{V} \downarrow.$$

Mass of a fixed volume vs. density of a fixed mass

These two phrasings are *not* the same, and ENGAA loves to exploit the difference (see Q4).

- **Density of a fixed mass** *decreases* on heating — correct, because the same mass now occupies more volume.
- **Mass of a fixed volume** does *not* increase — in fact, as the fluid expands, some of it leaves that fixed region, so the mass inside a fixed volume goes *down*, not up. Mass is conserved; it does not appear from nowhere on heating.

2.2 Convection Currents

Convection current

A convection current is the **bulk movement of a fluid driven by density differences**. Warmer, less-dense fluid rises; cooler, denser fluid sinks to take its place; the cycle repeats, transferring heat through the body of the fluid.

The mechanism, step by step (saucepan heated from below):

1. The heat source warms the water at the *bottom* first.
2. That water expands $\Rightarrow$ its volume increases $\Rightarrow$ its density decreases.
3. Being less dense than the cooler water above, the hot water **rises**.
4. Cooler, denser water **sinks** to fill the gap.
5. The sunk water is now over the heat source, warms, and the loop continues — this circulating loop is the convection current.

Everyday analogies for convection

- **Radiator (“heater”) in a room.** A radiator warms the air beside it; that air rises to the ceiling, cooler air is drawn along the floor to replace it, and a room-sized convection loop forms. That’s why radiators heat a whole room even though they only touch a little of the air.
- **Bridge expansion gaps.** Bridges are built with gaps (expansion joints) precisely because materials expand when heated. It’s the same underlying physics — heating → expansion — just in a solid, which is why engineers leave room for the deck to grow on a hot day. It is a vivid reminder that “hot ⇒ bigger”.

2.3 Gravitational Potential Energy (GPE)

Gravitational potential energy

$$E_{\text{GPE}} = mgh$$

m = mass (kg), g = gravitational field strength (= 10 N/kg on this paper), h = **vertical** height gained (m). Only the vertical rise counts — distance along a ramp does not, unless it is the vertical component.

This is the “useful” energy in any lifting/raising problem: the energy that actually ends up stored in the raised object. In both Q6 (car up a ramp) and Q12 (mass on a pulley), the *useful output* is exactly mgh .

2.4 Energy Conservation and Efficiency

Conservation of energy in a machine

Energy is never lost, only transferred. For any real machine:

$$\underbrace{E_{\text{input}}}_{\text{total in}} = \underbrace{E_{\text{useful}}}_{\text{what you wanted}} + \underbrace{E_{\text{wasted}}}_{\text{lost to surroundings}}$$

Rearranged, the wasted (“lost to surroundings”) energy is simply

$$E_{\text{wasted}} = E_{\text{input}} - E_{\text{useful}}.$$

Efficiency

$$\eta = \frac{\text{useful energy output}}{\text{total energy input}} \quad (\times 100\% \text{ if you want a percentage}).$$

Equivalently, useful output = $\eta \times$ input, and input = $\frac{\text{useful output}}{\eta}$.

The efficiency shortcut Tom drilled

For a “how much energy is wasted?” question you often **do not need the efficiency at all**. If you can find the useful energy directly (it’s mgh) and you are told the total input, then

$$E_{\text{wasted}} = E_{\text{input}} - E_{\text{useful}}$$

in one line. All that matters is *useful vs. total*. Don't get drawn into computing intermediate motor losses (see Q6).

2.5 Electrical Power and Energy

Electrical power and energy

$$P = IV \quad (\text{power} = \text{current} \times \text{voltage})$$

$$E = Pt = IVt \quad (\text{energy} = \text{power} \times \text{time})$$

P in watts (W), I in amps (A), V in volts (V), E in joules (J), t in seconds (s).

Rearranging $E = IVt$ to make the current the subject gives the form used in Q12:

$$I = \frac{E}{Vt}.$$

This is the electrical energy *delivered by the supply*, so if the system is not perfectly efficient, the E here is the *input* energy, not the useful mgh .

2.6 Nuclear Decay — Alpha and Beta

The two decays you must know cold

Write a nuclide as ${}^A_Z\text{X}$: A = mass number (nucleons), Z = atomic number (protons).

- **Alpha (α) decay** — emits a helium nucleus ${}^4_2\text{He}$:

$$\Delta A = -4, \quad \Delta Z = -2.$$

- **Beta-minus (β^-) decay** — a neutron turns into a proton, emitting an electron ${}^0_{-1}\text{e}$:

$$\Delta A = 0, \quad \Delta Z = +1.$$

Gamma (γ) is electromagnetic radiation released from the nucleus; it changes neither A nor Z .

The one fact that cracks decay-chain questions

Only alpha decay changes the mass number. Beta leaves A untouched. So in any decay chain, the mass number can only ever sit at its starting value or drop in steps of 4 — the number of alphas fixes the mass number completely. Read the mass number *first*, then worry about the atomic number. This is exactly the method that solves Q14 in seconds.

2.7 Nuclear Equations — Conservation

Balancing a nuclear equation

In *any* nuclear reaction, two quantities are conserved:

- **Mass number A** (top numbers): sum on the left = sum on the right.
- **Atomic number Z** (bottom numbers): sum on the left = sum on the right.

Treat any unknowns as algebra and write down the two balance equations. A free neutron is ${}^1_0\text{n}$: it carries mass number 1 but atomic number 0, so it counts in the A balance but contributes nothing to the Z balance.

2.8 Kinematics — Distance from a Velocity–Time Graph

Area under a v – t graph = distance

On a velocity–time graph, the **area between the line and the time axis is the distance travelled**. Break the area into simple shapes:

- **Rectangle** (constant velocity): $\text{area} = v \times t$.
- **Triangle** (starting or ending at zero): $\text{area} = \frac{1}{2} b h$.
- **Trapezium** (velocity changes linearly between two non-zero values): $\text{area} = \frac{u + v}{2} \times t$,
i.e. average velocity $\times$ time.

Trapezium, not rectangle

When a car decelerates from u to a *non-zero* v , the region under that part of the graph is a **trapezium**, not a rectangle. Using $v \times t$ with just one velocity is the classic error. Always: *identify the section of the graph the question asks about first, then decide the shape whose area you need.*

3 Question-by-Question Breakdown

Each question below records the full setup, the exact options, Tom’s method, the answer, every mistake made and corrected, and the coaching and analogies Tom used.

3.1 Q1 — Opening Maths Warm-up

Topic: algebra (quadratic inequality) — the style of the opening maths question that begins the paper.

Tom’s feedback on Q1 was that it was “a straightforward maths problem and caused no issue,” and that it was “a useful warm-up because these early questions often need to be completed quickly so that you have time for the more physics-heavy questions later in the paper.” The representative early-maths question worked at the board was a quadratic inequality:

Solve fully the inequality $2x^2 + x \geq 15$

Options: **A** $x \leq -3$ or $x \geq 2.5$ **B** $x \leq -2.5$ or $x \geq 3$ **C** $-3 \leq x \leq 2.5$ **D** $-2.5 \leq x \leq 3$

E $x \leq -1.5$ or $x \geq 5$ **F** $x \leq -5$ or $x \geq 1.5$ **G** $-5 \leq x \leq 1.5$ **H** $-1.5 \leq x \leq 5$

Method (Tom’s three steps):

1. **Make it equal to zero:** $2x^2 + x - 15 \geq 0$.
2. **Find the roots** by factorising: $(2x - 5)(x + 3) \geq 0 \Rightarrow x = 2.5$ or $x = -3$.
3. **Sketch** the upward parabola cutting the axis at -3 and 2.5 . The curve is ≥ 0 *outside* the roots.

$\Rightarrow x \leq -3$ or $x \geq 2.5$. Answer **A**.

Warm-up questions are speed questions

Do not linger on the opening maths. These marks are there to be banked fast so your time budget survives for the physics later in the paper. If a quadratic inequality doesn’t factorise cleanly, that’s a signal you have mis-copied it — ENGAA numbers are always tidy.

3.2 Q2 — Velocity–Time Graph (Distance while Decelerating)

Topic: kinematics — area under a v – t graph.

Setup

A velocity–time graph shows part of a car’s journey (velocity in m s^{-1} , time in s):

- 0–25 s: accelerates uniformly from 0 to 30 m/s.
- 25–50 s: constant at 30 m/s.
- 50–110 s: **decelerates** uniformly from 30 to 10 m/s.
- 110–150 s: constant at 10 m/s.

Question: What is the total distance the car travels *while it is decelerating*?

Options: **A** 400 m **B** 500 m **C** 550 m **D** 600 m **E** 700 m **F** 1100 m **G** 1400 m **H** 1800 m

Tom’s method. The key is *careful reading*: the question asks only for the distance during the **deceleration** section (50–110 s), not the whole journey. Distance is the area under that

section. Because the velocity falls from 30 to 10 (both non-zero), the shape is a **trapezium**:

$$S = \frac{u+v}{2} \times t = \frac{30+10}{2} \times (110-50) = \frac{40}{2} \times 60 = 20 \times 60 = 1200 \text{ m.}$$

So the distance travelled while decelerating is 1200 m.

Phuc's Mistake & Correction

On the board the deceleration area was first mis-evaluated: the trapezium was set up correctly as $\frac{30+10}{2} \times (110-50)$ but the arithmetic slipped to “500 m”. Multiplying properly, $\frac{40}{2} \times 60 = 20 \times 60 = 1200 \text{ m}$. **Correction:** take the average of the two velocities (20) and multiply by the *time interval* (60 s), not by a single velocity.

Read what is actually asked

Two traps stacked in one question:

1. Computing the *whole* journey's distance instead of only the decelerating part.
2. Treating the decelerating region as a rectangle. It is a trapezium.

Tom: “always identify the section of the graph first, then decide the shape whose area you need.”

3.3 Q4 — Convection Currents

Topic: thermal physics — density, expansion, convection.

Setup

When a saucepan of water is heated from below, convection currents form and transfer heat through the liquid. Three statements about the water as it is heated:

1. The mass of a fixed volume of the water increases.
2. The density of a fixed mass of the water decreases.
3. The volume of a fixed mass of the water increases.

Question: Which of these statements help(s) to explain how convection currents are formed?

Options: A none B 1 only C 2 only D 3 only E 1 and 2 F 1 and 3 G 2 and 3 H 1, 2 and 3

Tom's reasoning from first principles. Heat the bottom of the pan $\rightarrow$ that water gets hotter $\rightarrow$ it **expands**, so its **volume increases** (statement 3, $\checkmark$). For a fixed mass, $\rho = m/V$ with V up means **density decreases** (statement 2, $\checkmark$). The less-dense hot water rises, cooler denser water sinks to replace it — and that circulation *is* the convection current. Statement 1 is wrong: heating does not add mass, so the “mass of a fixed volume” does not increase (if anything, expanding fluid leaves the fixed volume, so that mass falls).

$\Rightarrow$ statements 2 and 3 only. Answer **G**.

Phuc's Mistake & Correction

Phuc correctly picked statements 2 and 3 (answer G). The subtlety Tom nailed down was the *direction of causation*: it is the **volume increasing that causes the density to decrease**, not the other way round. Heating $\rightarrow$ expansion $\rightarrow V \uparrow \rightarrow \rho \downarrow$. Get the causal chain in that order and statement 1 is obviously the odd one out.

- 4 When a saucepan of water is heated from below, convection currents form and transfer heat through the liquid.

Here are three statements about the water as it is heated:

- ☒ 1 The mass of a fixed volume of the water increases.
☒ 2 The density of a fixed mass of the water decreases.
☒ 3 The volume of a fixed mass of the water increases.

Which of these statements help(s) to explain how convection currents are formed?

- A none of them
 B 1 only
 C 2 only
 D 3 only
 E 1 and 2 only
 F 1 and 3 only
☒ G 2 and 3 only
 H 1, 2 and 3

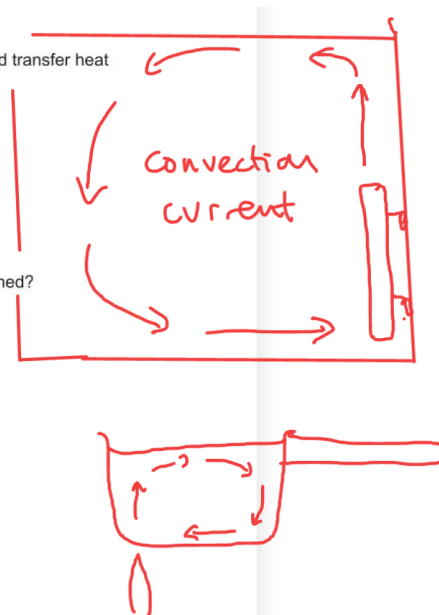


Figure 1: *

Board (img_01): statement 1 crossed out, statements 2 and 3 ticked; convection-current loop and saucepan sketch; answer G circled.

Watch the exact wording “fixed mass” vs “fixed volume”

The whole question hinges on which quantity is held fixed. “Density of a fixed *mass*” and “mass of a fixed *volume*” sound similar but behave oppositely under heating. Underline the held-fixed quantity before you judge each statement.

3.4 Q6 — Useful and Wasted Energy (Car up a Ramp)

Topic: energy conservation and efficiency.

Setup

An electric motor pulls a broken-down car slowly up a ramp onto a breakdown truck. Mass $m = 1200$ kg, vertical height $h = 1.0$ m. The total input energy to the motor is 28 kJ and it is 75% efficient. Energy is lost to the surroundings from the motor and other causes. ($g = 10$ N/kg.)

Question: What is the **total** energy lost to the surroundings?

Options: A 7.0 kJ B 9.0 kJ C 12 kJ D 16 kJ E 21 kJ F 28 kJ G 33 kJ

The fast route. Useful output = GPE gained by the car:

$$E_{\text{useful}} = mgh = 1200 \times 10 \times 1.0 = 12\,000 \text{ J} = 12 \text{ kJ.}$$

Total input = 28 kJ. Therefore

$$E_{\text{wasted}} = E_{\text{input}} - E_{\text{useful}} = 28 - 12 = 16 \text{ kJ.}$$

Answer **D**.

6 An electric motor is used to pull a broken-down car slowly from the road up a ramp on to the back of a breakdown truck.

The car has a mass of 1200 kg and is lifted through a vertical height of 1.0 m.

The total input energy to the motor is 28 kJ and it is 75% efficient.

In the process of lifting the car, energy is lost to the surroundings from the motor and from other causes.

What is the **total** energy lost to the surroundings?

total energy = useful energy + wasted energy

(gravitational field strength = 10 N kg^{-1})

A 7.0 kJ
B 9.0 kJ
C 12 kJ
D 16 kJ
E 21 kJ
F 33 kJ

28000 * 0.75 = 21000 for useful work
and lost 28000 * 0.25 = 7000
21000 - 12000 = 9000
7000 + 9000 = 16000 J $\Rightarrow$ D

Figure 2: *

Board (img_02): energy-flow arrows 28 kJ in $\rightarrow$ 7 kJ (motor loss) + 21 kJ (motor output) $\rightarrow$ 9 kJ (further loss) + 12 kJ (useful GPE); “total energy = useful energy + wasted energy”; answer D.

Phuc’s Mistake & Correction

Phuc tried to route the whole thing *through* the 75% efficiency figure: computing $0.75 \times 28 = 21 \text{ kJ}$ as the motor’s useful output and $0.25 \times 28 = 7 \text{ kJ}$ as the motor’s internal loss, then trying to reconcile that 21 kJ against the 12 kJ actually lifted. **Tom’s correction:** for “total energy lost to the surroundings” you never need the 75% at all. The only two numbers that matter are the *useful* energy ($mgh = 12 \text{ kJ}$) and the *total input* (28 kJ); the loss is their difference, 16 kJ.

“All that matters is useful vs total”

The 75% is a distractor for *this* question. If you can see the useful energy directly, subtract it from the input and stop. The board even confirms the two-stage breakdown is self-consistent — motor loss 7 kJ, then $21 - 12 = 9 \text{ kJ}$ lost after the motor, and $7 + 9 = 16 \text{ kJ}$ — but that longer route is exactly the “unnecessary detail about the motor” Tom told you to skip. Spot the useful energy, subtract, done.

3.5 Q8 — Gamma-Ray Path Difference (PET Scanner)

Topic: waves / kinematics of light — $d = v \times t$ and path difference.

Setup

In a medical scanner (Positron Emission Tomography, PET) a source inside the patient emits pairs of gamma-rays simultaneously in opposite directions towards two detectors. The detectors are 3.0 m apart. When the source is at Q , half-way between the detectors, the two gamma-rays arrive at the same time. In a scan the gamma-rays arrive with a time difference of $4.0 \times 10^{-10} \text{ s}$. The gamma-rays travel at $3.0 \times 10^8 \text{ m/s}$.

Question: How far from Q is the source?

Options: A 6.0 mm B 12 mm C 24 mm D 6.0 cm E 12 cm F 24 cm

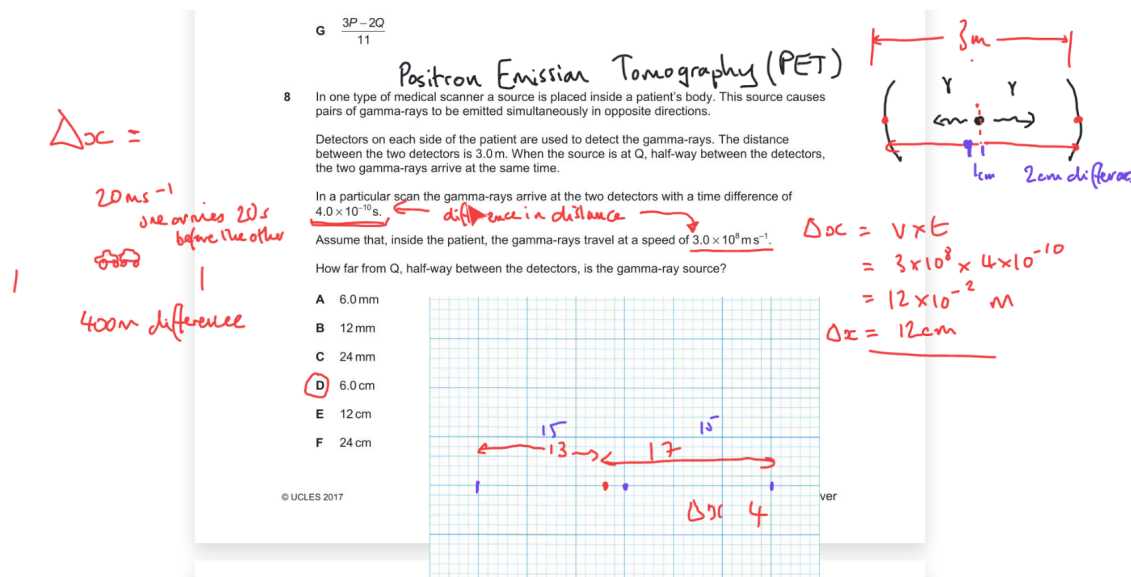


Figure 3: *

Board (img_03): PET diagram, the two-cars analogy (“20 m/s, one arrives 20 s before the other $\Rightarrow$ 400 m difference”), grid-paper model (15/15 $\rightarrow$ 13/17, $\Delta x = 4$), and $\Delta x = v \times t = 3 \times 10^8 \times 4 \times 10^{-10} = 12 \times 10^{-2}$ m = 12 cm; answer D circled.

Step 1 — path difference from the time delay. Both rays travel at the same speed, so the extra distance one covers is

$$\Delta x = v \times t = (3.0 \times 10^8) \times (4.0 \times 10^{-10}) = 12 \times 10^{-2} \text{ m} = 12 \text{ cm}.$$

Step 2 — the key geometry: path difference = $2x$. Let the source sit a distance x from the midpoint Q . Then one path becomes x longer and the other becomes x shorter, so the difference between the two path lengths is

$$(\text{longer}) - (\text{shorter}) = (d + x) - (d - x) = 2x.$$

Hence $2x = 12$ cm, giving $x = 6.0$ cm. Answer **D**.

Phuc’s Mistake & Correction

Phuc computed $\Delta x = 12$ cm and was ready to give 12 cm as the answer. **Tom’s correction:** 12 cm is the *path difference* between the two rays, not the displacement of the source. Because shifting the source by x makes one path longer *and* the other shorter, the path difference is $2x$ — so the displacement is *half* of 12 cm, i.e. 6.0 cm. Phuc found the “ $\times 2$ ” relationship genuinely tricky, so Tom demonstrated it two ways.

Analogy 1 — the two cars

Picture two cars both driving at the same speed, 20 m/s. If one arrives 20 s before the other, the difference in the distances they covered is $v \times t = 20 \times 20 = 400$ m. Exactly the same logic turns the gamma-ray time delay into a *path difference*: same speed, different arrival times $\Rightarrow$ difference in distance = $v \times \Delta t$.

Analogy 2 — the grid-paper demo

On grid paper Tom put the source at the midpoint: each path is 15 squares, so the difference is 0. Now slide the source 2 squares towards one detector: that path becomes 13, the other becomes 17. The path difference is $17 - 13 = 4$ squares — *twice* the 2-square shift. This is the “ $2x$ ” relationship made concrete: **move the source by x , change the path difference by $2x$** , so displacement $= \frac{1}{2} \times$ path difference.

Which information actually matters?

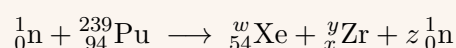
The 3.0 m detector separation is a **distractor** — it never enters the calculation. All you need is the speed and the time difference to get the path difference, then halve it. Recognising that the detector gap is irrelevant is exactly the “decide which information matters” skill Tom is training.

3.6 Q10 — Balancing a Nuclear Fission Equation

Topic: nuclear physics — conservation of mass and atomic number.

Setup

A plutonium-239 nucleus absorbs a neutron and undergoes fission, producing xenon and zirconium. The equation has some non-zero integers replaced by w, x, y, z :



Question: Which equation is correct?

Options: A $w + y = 240$ B $z = 240 - (w + y)$ C $x = 40 - z$

D $94 = 54 + x + 1$ E $240 = 54 + x$ F $94 = w + y + 1$

11

10 When a plutonium-239 nucleus absorbs a neutron it undergoes nuclear fission. One particular fission reaction results in the creation of xenon and zirconium as daughter nuclei. The nuclear equation for this reaction is shown but with some non-zero integers replaced by the letters w, x, y, z .

$${}_0^1\text{n} + {}_{94}^{239}\text{Pu} \rightarrow {}_{54}^w\text{Xe} + {}_x^y\text{Zr} + z{}_0^1\text{n}$$

Which equation is correct?

A $w + y = 240$
 B $z = 240 - (w + y)$
 C $x = 40 - z$
 D $94 = 54 + x + 1$
 E $240 = 54 + x$
 F $94 = w + y + 1$

Handwritten notes:
 $94 = 54 + x \Rightarrow x = 40$
 $w + y + z = 240$
 ✓ B) $z = 240 - (w + y)$
 C) miss variables
 D, E's x isn't 40, not correct
 F is incorrect because missing z

Figure 4: *

Board (img_04): $94 = 54 + x \Rightarrow x = 40$; $w + y + z = 240$; option B ticked; C “miss variables”; D, E’s x isn’t 40; F incorrect because it misses z .

Method. Apply the two conservation laws.

- **Mass number** (tops): $1 + 239 = w + y + z$, so

$$w + y + z = 240.$$

- **Atomic number** (bottoms): $0 + 94 = 54 + x + 0$, so

$$x = 40.$$

(The z neutrons carry atomic number 0, so z does not appear here.)

Now test the options against these facts. The mass-number law rearranges to $z = 240 - (w + y)$, which is exactly option **B**. The board notes why the rest fail:

- **C** ($x = 40 - z$): wrong — $x = 40$ outright; z has nothing to do with x (“miss variables”).
- **D** ($94 = 54 + x + 1$) and **E** ($240 = 54 + x$): both force $x \neq 40$, contradicting the atomic-number balance.
- **F** ($94 = w + y + 1$): wrong — it drops z and mixes mass numbers (w, y) into an atomic-number statement.

Answer **B**.

Phuc’s Mistake & Correction

Phuc’s instinct was to write the mass-number balance as $w + y + z = 240$ (correct) but momentarily wanted to include z in the *atomic*-number balance. Tom’s correction: z counts free neutrons, and a neutron’s atomic number is 0, so it drops out of the bottom-row balance entirely — the atomic-number equation is just $94 = 54 + x$, giving $x = 40$.

Treat the unknowns as algebra and check every option

Write the two balance equations first, then run down the options eliminating any that use a variable in the wrong row or drop a term. The correct answer is often a *rearrangement* of a balance law rather than the law in its most obvious form — here $z = 240 - (w + y)$ rather than $w + y + z = 240$.

3.7 Q12 — Motor, Pulley and Efficiency (Current)

Topic: electrical power/energy with efficiency.

Setup

An electric motor on a constant 12 V d.c. supply lifts a mass of 20 kg with a rope and pulley, raising it vertically through 6.0 m in a time of 5.0 s. The complete lifting system (motor, rope, pulley) is 80% efficient. ($g = 10 \text{ N/kg}$.)

Question: What is the current in the electric motor?

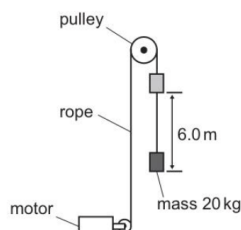
Options: **A** 1.6 A **B** 2.0 A **C** 2.5 A **D** 16 A **E** 20 A **F** 25 A

Method.

1. **Useful output** $= mgh = 20 \times 10 \times 6.0 = 1200 \text{ J}$.
2. If the system were 100% efficient, the supply would deliver this same 1200 J, and the current would be

$$I = \frac{E}{Vt} = \frac{1200}{12 \times 5} = 20 \text{ A}.$$

- 12 An electric motor is connected to a constant 12 V d.c. supply. The motor is used to lift a mass of 20 kg by means of a rope and pulley. The mass is lifted vertically through a height of 6.0 m in a time of 5.0 s. The complete lifting system (motor, rope and pulley) is 80% efficient.



What is the current in the electric motor?

(gravitational field strength = 10 N kg^{-1})

- A 1.6 A
- B 2.0 A
- C 2.5 A
- D 16 A
- E 20 A
- F 25 A

$$V = 12 \text{ V}$$

$$W = mgh = 20 \times 10 \times 6 = 1200$$

$$I = E/(Vt) = 1200 / (12 \times 5) = 20 \text{ A}$$

$$0.8 \times \text{input} = 20$$

$$\Rightarrow \text{input} = 20/0.8 = 25 \Rightarrow \text{F}$$

$$P = IV \quad E = P \times t$$

$$E = IVt$$

Figure 5: *

Board (img_05): pulley/motor diagram, $P = IV$, $E = IVt$; $W = mgh = 20 \times 10 \times 6 = 1200$;
 $I = E/(Vt) = 1200/(12 \times 5) = 20 \text{ A}$; then $0.8 \times \text{input} = 20 \Rightarrow \text{input} = 20/0.8 = 25 \Rightarrow \text{F}$.

3. But the system is only 80% efficient, so the **input** energy must be *larger* than the useful output. The current the supply actually draws scales up by $1/\eta$:

$$I_{\text{input}} = \frac{20}{0.8} = 25 \text{ A}.$$

Answer **F**.

Phuc's Mistake & Correction

Phuc's method was completely correct up to the current: he found $mgh = 1200 \text{ J}$ and $I = 1200/(12 \times 5) = 20 \text{ A}$ — and stopped there, giving **E (20 A)**. He **forgot the 80% efficiency factor**. **Tom's words**: “everything you did was right, you just missed the last bit.” Because the motor is not perfect, the electrical input power (and hence the current from the 12 V supply) must exceed the useful value: divide by 0.8 to get 25 A. The correct answer is **F (25 A)**, not E.

Efficiency always makes the input BIGGER

When you convert useful output into required input, you *divide* by the efficiency (input = useful/ η), which *increases* the number. A quick sanity check: an 80%-efficient machine must be fed *more* energy than it usefully delivers, so the input current 25 A must be larger than the “ideal” 20 A — never smaller. If your efficiency step made the number go down, you divided the wrong way.

3.8 Q14 — Alpha/Beta Decay Chain (Impossible Nuclide)

Topic: nuclear physics — decay sequences.

Setup

The nuclide P_QX decays to the stable nuclide Y . During this process four particles are emitted: one α -particle and three β^- particles.

Question: Which of the following is **not** a nuclide that could be formed at any stage during this process?

opt	mass	atomic	opt	mass	atomic
A	P	$Q - 1$	E	$P - 4$	$Q - 2$
B	P	$Q + 1$	F	$P - 4$	$Q - 1$
C	P	$Q + 2$	G	$P - 4$	Q
D	P	$Q + 3$	H	$P - 4$	$Q + 1$

Handwritten notes at the top show two decay paths:

$${}^P_QX \rightarrow {}^4_2\alpha + {}^{P-4}_{Q-2}Y$$

$${}^P_QX \rightarrow {}^0_{-1}\beta + {}^P_{Q+1}Y$$

14 The nuclide P_QX decays to the stable nuclide Y . During this process four particles are emitted: an α -particle and three β^- particles.

Which of the following is **not** a nuclide that could be formed at any stage during this process?

nuclide	atomic mass	atomic number
A	P	$Q - 1$
B	P	$Q + 1$
C	P	$Q + 2$
D	P	$Q + 3$
E	$P - 4$	$Q - 2$
F	$P - 4$	$Q - 1$
G	$P - 4$	Q
H	$P - 4$	$Q + 1$

Handwritten notes next to the table:

- Next to A: 1β
- Next to B: 2β
- Next to C: 3β
- Next to D: 1α
- Next to E: 1α
- Next to F: 1α
- Next to G: 1α
- Next to H: 1α

Tom's note: "atomic number is down - so need an alpha but can't have an alpha and have the same mass."

Tom

$P - 4$
 $Q + 1$

Figure 6: *

Board (img_06): α gives $A - 4$, $Z - 2$; β^- gives A , $Z + 1$; the reachable states listed for 0/1/2/3 betas before and after the alpha; Tom's note "atomic number is down — so need an alpha, but can't have an alpha and keep the same mass." Answer A.

Tom's fast method — read the mass number first. The mass number can *only* be changed by the alpha decay (β^- leaves it fixed). So at any stage the mass number is either:

- **still P** — the alpha has not happened yet, so *only betas* have occurred. Betas raise Z , so the atomic number can only be Q , $Q + 1$, $Q + 2$, $Q + 3$ (0 to 3 betas). *It can never be below Q .*
- **$P - 4$** — the alpha has happened (Z dropped by 2), plus 0–3 betas (Z up by 0–3), giving atomic numbers $Q - 2$, $Q - 1$, Q , $Q + 1$.

Now scan the options. Options B, C, D (P with $Q + 1$, $Q + 2$, $Q + 3$) are all reachable before the alpha. Options E, F, G, H ($P - 4$ with $Q - 2$, $Q - 1$, Q , $Q + 1$) are all reachable after it. **Option A is** (P , $Q - 1$): mass still P but atomic number *below* Q . To drop the atomic number you need an alpha — but an alpha would also drop the mass to $P - 4$. You cannot have a reduced atomic number *and* the original mass number. So $(P, Q - 1)$ can never occur.

Answer **A**.

The two-line reasoning Tom wrote on the board

“Atomic number is down, so you need an alpha — but you can’t have an alpha and still have the same mass.” That single sentence is the whole solution. Whenever a decay-chain option shows the *original* mass number but a *lowered* atomic number, it is impossible, because the only thing that lowers Z (the alpha) also lowers A .

General technique for “which nuclide is impossible”

1. Fix the mass number first: it can only be P (no alpha yet) or $P - 4$ (alpha done). Any other mass number is instantly impossible.
2. For each allowed mass number, list the range of atomic numbers the betas can produce.
3. The impossible option is the one whose (A, Z) pair falls outside both reachable sets.

4 Tom's Written Feedback (Verbatim)

The following is Tom's full written feedback, reproduced verbatim from his Slack post after the lesson.

DETAILED FEEDBACK	
Overview	This was a good first ESAT preparation session. You worked through the opening maths question and then focused on the physics questions from ENGAA 2017, rather than trying to complete every question in order. This is a sensible approach for ESAT preparation because the aim is not just to get answers, but to build the habit of recognising the physics quickly and choosing the most efficient route. The main theme was careful reading of the question. Several of these questions had a short route once the context was understood: area under a velocity-time graph, density changes in convection, useful and wasted energy, gamma-ray path difference, conservation in nuclear equations, and efficiency in a motor system. These are exactly the kinds of ideas that can come up in admissions-style tests.
Question feedback	Q1 was a straightforward maths problem and caused no issue. It was a useful warm-up because these early questions often need to be completed quickly so that you have time for the more physics-heavy questions later in the paper. Q2 was a velocity-time graph question. The key point was to read the wording carefully: it asked only for the distance travelled while the car was decelerating. Distance from a velocity-time graph is the area under the graph, and for this section the area was a trapezium rather than a rectangle. This is a common source of errors, so always identify the section of the graph first, then decide the shape whose area you need. Q4 was about convection currents in water heated from below. The physics needed here was that heating causes a fixed mass of water to expand, so its volume increases and its density decreases. The less dense hot water rises, while cooler, denser water sinks to replace it. That movement of fluid sets up the convection current. The important distinction is that the mass of a fixed volume does not increase; the useful explanation is based on expansion and lower density.
Independent work	Q6 was an energy question about raising a car onto a breakdown truck. The useful work done on the car is the gain in gravitational potential energy, mgh, which is $1200 \times 10 \times 1.0 = 12 \text{ kJ}$. The total energy input is 28 kJ, so the energy lost to the surroundings is $28 \text{ kJ} - 12 \text{ kJ} = 16 \text{ kJ}$. You can solve this quickly by spotting the useful energy directly, rather than getting drawn into unnecessary detail about the motor. Q8 was the gamma detector question. The context matters: a source emits two gamma rays in opposite directions towards two detectors. The time delay gives the difference in path length using distance = speed $\times$ time. The subtle point is that moving the source a distance x from the midpoint makes one path x longer and the other x shorter, so the path difference is $2x$. This is why the displacement of the source is half the path difference, giving 6 cm. Q10 was balancing a nuclear equation using conservation of nucleon number and proton number. In nuclear equations, the total mass number must balance on both sides and the total atomic number must also balance on both sides. Treating the unknowns algebraically is a good strategy here. The correct option was B because the missing neutrons must account for the difference between the initial total nucleon number and the total already in the daughter nuclei. Q12 was the pulley and motor question. Your method was right: the useful output energy is the gain in gravitational potential energy, mgh, and because it happens in a time you can connect this to power. The important missed step was the 80% efficiency factor. Since the system is not 100% efficient, the electrical input power must be larger than the useful output power, so the current needed from the 12 V supply is larger than the value found from useful power alone.
Next steps	For independent work, continue with the ENGAA 2017 paper but keep the same focus: identify the physics, write down the key idea, and then look for the quickest route. It would be particularly useful to redo Q2, Q4, Q6, Q8, Q10 and Q12 without notes, checking that you can explain the physics as well as get the answer. We also discussed setting you a A level paper to help assess your understanding. I will send one of these. Next time we can continue with more ESAT-style questions and keep building speed. The priority is to become more confident at spotting the topic quickly, deciding which information in the question actually matters, and avoiding long methods when a simple physical argument or proportional shortcut will do the job.

Figure 7: *

Tom's written feedback as posted (*img_08* and *img_09*).

Overview

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Question feedback

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Independent work & next steps (verbatim)

“For independent work, continue with the ENGAA 2017 paper but keep the same focus: identify the physics, write down the key idea, and then look for the quickest route. It would be particularly useful to redo Q2, Q4, Q6, Q8, Q10 and Q12 without notes, checking that you can explain the physics as well as get the answer. We also discussed setting you a A level paper to help assess your understanding, I will send one of these.

Next time we can continue with more ESAT-style questions and keep building speed. The priority is to become more confident at spotting the topic quickly, deciding which information in the question actually matters, and avoiding long methods when a simple physical argument or proportional shortcut will do the job.”

Tom’s covering message with the A-level paper

“Here is a Unit 1 paper to work through — if you can do this on your iPad and edit on screen that will be easier for me to mark. Take as long as you need and use your notes if needed — just avoid the mark scheme or getting AI help as it will get you to the answer too easily.”

5 What Tom Asked You to Do Next

Your action list

1. **Redo Q2, Q4, Q6, Q8, Q10 and Q12 *without notes*.** Check that you can *explain the physics* as well as get the answer — not just reproduce the arithmetic. These are the six that had a mistake, a subtlety, or a shortcut worth internalising.
2. **Continue the ENGAA 2017 paper.** Keep the same focus for every question: *identify the physics* → *write down the key idea* → *find the quickest route*.
3. **Do the Unit 1 A-level Physics past paper Tom is sending.** Work it on your iPad and annotate on screen so Tom can mark it easily. Take as long as you need and use notes if needed, but **avoid the mark scheme and avoid AI help** — they short-circuit the learning. This paper is a diagnostic of your underlying understanding.
4. **Share your ESAT topic-mapping spreadsheet with Tom** so he can see which topics you have covered and target future sessions.

Lesson-planning notes

- **Next week:** Tom is available most days — book a follow-up session.
- **After that:** Tom is on holiday for roughly two weeks, so plan the next couple of sessions before that break.
- **Direction of future sessions:** more ESAT-style questions, building speed, and the A-level Unit 1 diagnostic to calibrate depth of understanding.

How to redo the six questions well

Don't just re-read the solutions here. For each of Q2, Q4, Q6, Q8, Q10, Q12: cover the notes, write the *one key idea* from memory, solve it cold, and only then compare against this guide. The value is in reproducing the *reasoning*, especially the three “ $\times 2$ / $\div 0.8$ / read-the-question” traps that caught you the first time.

6 Must-Memorise Formula Reference

Every relationship used in this lesson, in one place. No formula sheet is provided in the ESAT — these must be automatic.

Quantity	Formula	Notes / used in
Density	$\rho = \frac{m}{V}$	Fixed mass: $V \uparrow \Rightarrow \rho \downarrow$. (Q4)
Thermal expansion	heat $\Rightarrow V \uparrow \Rightarrow \rho \downarrow$	Drives convection; mass conserved. (Q4)
Gravitational PE	$E = mgh$	Useful lifting energy; h is vertical. Use $g = 10$. (Q6, Q12)
Energy conservation	$E_{\text{in}} = E_{\text{useful}} + E_{\text{wasted}}$	$E_{\text{wasted}} = E_{\text{in}} - E_{\text{useful}}$. (Q6)
Efficiency	$\eta = \frac{E_{\text{useful}}}{E_{\text{in}}}$	Input = E_{useful}/η (bigger!). (Q12)
Electrical power	$P = IV$	Watts = amps $\times$ volts. (Q12)
Electrical energy	$E = IVt, I = \frac{E}{Vt}$	Use <i>input</i> energy when $\eta < 1$. (Q12)
Distance (const. v)	$d = vt$	Also path difference = $v \Delta t$. (Q2, Q8)
Area under $v-t$	distance = area	Rectangle / triangle / trapezium. (Q2)
Trapezium area	$\frac{u+v}{2} \times t$	Average velocity $\times$ time. (Q2)
Path difference	$\Delta x = 2x$	Shift source by $x \Rightarrow$ path diff $2x$. (Q8)
Alpha decay	$\Delta A = -4, \Delta Z = -2$	Emits ${}^4_2\text{He}$. (Q14)
Beta ⁻ decay	$\Delta A = 0, \Delta Z = +1$	Emits ${}^0_{-1}\text{e}$. (Q14)
Gamma decay	$\Delta A = 0, \Delta Z = 0$	EM radiation only.
Nuclear balance	$\sum A, \sum Z$ conserved	Neutron = mass 1, charge 0. (Q10, Q14)

Q	Topic	Answer	One-line takeaway
1	Quadratic inequality	A	Rearrange to $= 0$, factorise, sketch, read outside roots.
2	v - t graph	F (1200 m)	Decelerating section only; trapezium, not rectangle.
4	Convection	G	Heat $\rightarrow V \uparrow \rightarrow \rho \downarrow$; statements 2 & 3.
6	Useful/wasted energy	D (16 kJ)	$E_{\text{wasted}} = 28 - mgh = 28 - 12$.
8	Gamma path diff.	D (6.0 cm)	$\Delta x = vt = 12$ cm; displacement $= \Delta x/2$.
10	Nuclear balance	B	$z = 240 - (w + y)$; $x = 40$ from Z -balance.
12	Motor efficiency	F (25 A)	$I = mgh/(Vt) = 20$, then $\div 0.8 = 25$.
14	Decay chain	A	$(P, Q-1)$ impossible: lower Z needs alpha, but alpha lowers A .

*Identify the physics first. Approximate freely. Use $g = 10$. Bank the easy marks.
 Recognise $\rightarrow$ shortcut $\rightarrow$ answer. — End of guide —*